

Sprint 3 Report

CSE 4316

Ada Lovelace

November 3rd, 2026

Sprint start date:	October 20th, 2026
Sprint end date:	November 3rd, 2026
Project title:	TerraScout Autonomous Ground Rover
Team name:	Team Trailblazer
Partners:	Alan Turing Grace Hopper John Von Neumann Charles Babbage
Instructor:	Christopher D. McMurrough

1 Sprint Goal

Stand up the perception pipeline: synchronized RGB/thermal capture tagged with pose, and a first fault-detection baseline evaluated against the sponsor validation set.

2 Sprint Backlog

Breakdown of planned tasks assigned to the team (work units in hours).

Task description	Estimated work	Actual work
Integrate thermal camera driver (ROS 2 node)	6.0	8.0
Hardware trigger for RGB/thermal sync	4.0	5.0
Pose stamping of captured frames	3.0	2.5
Label sponsor validation set	5.0	4.0
Train MobileNet baseline detector	6.0	7.0
Evaluate detector (precision/recall)	3.0	3.0
Ground-station telemetry stub	4.0	3.5
TOTAL	31.0	33.0

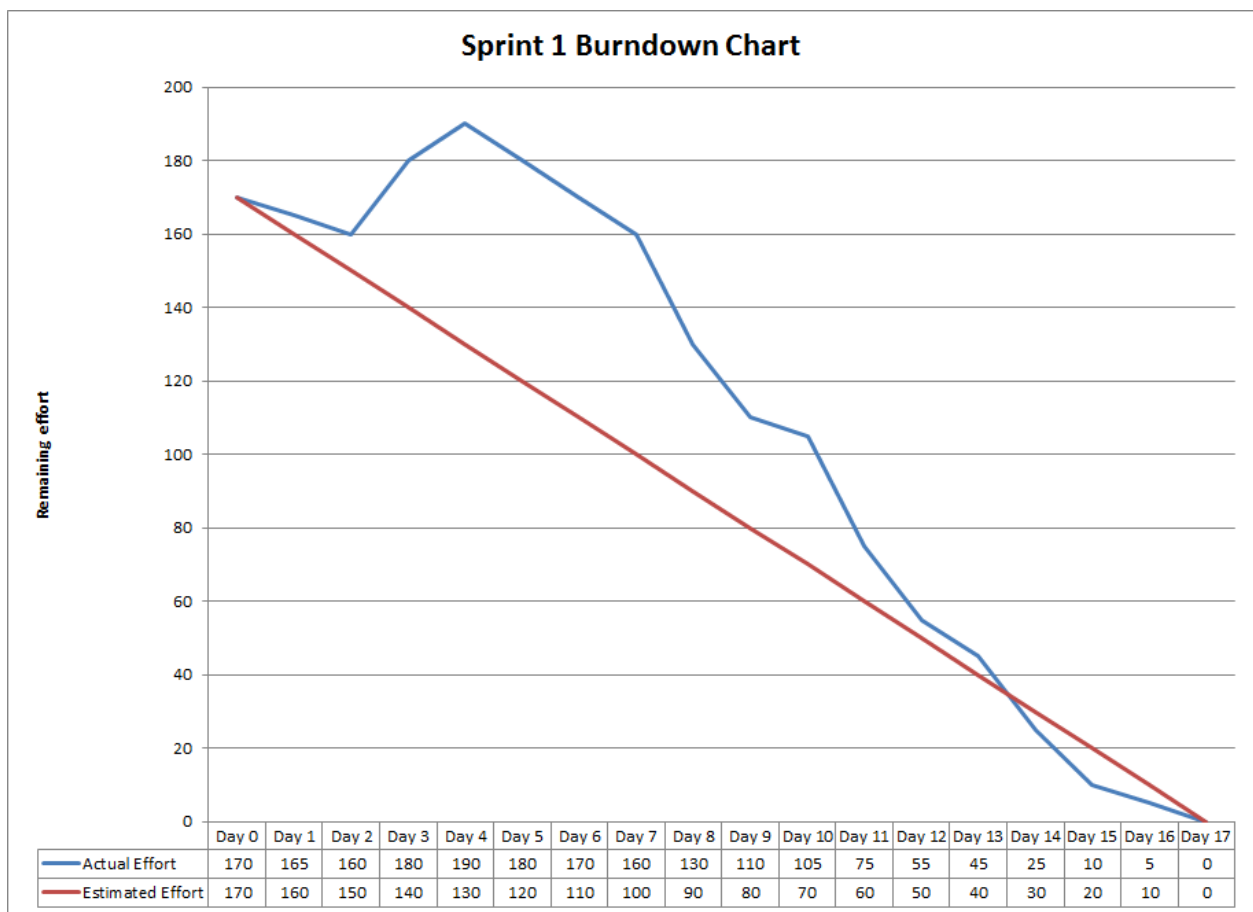
3 Individual Time Expenditures

Summary of tasks performed by the individual (hours).

Task Description	Actual work	Percent complete
Label sponsor validation set	4.0	100%
Train MobileNet baseline detector	7.0	100%
Evaluate detector (precision/recall)	3.0	100%
Investigate thermal false positives	2.0	60%
TOTAL	16.0	-

4 Team Burndown Chart

Ideal (baseline) and actual daily team effort are shown below.



5 Individual Retrospective

The following describe actions to start, stop, or continue next sprint.

Actions to **start** doing (top 3)

- Version-control trained model artifacts with dataset hashes for reproducibility.
- Write the evaluation script first, before training, to fix the metric.
- Book the outdoor test slot at the start of the sprint, not the end.

Actions to **stop** doing (top 3)

- Hand-tuning thresholds in a notebook instead of in the node config.
- Committing large raw thermal captures to Git (use the shared drive).
- Estimating CV tasks without a data-labeling sub-estimate.

Actions to **continue** doing (top 3)

- Pairing with John on the pose-stamping interface.
- Daily short syncs on integration blockers.
- Keeping the detector behind a stable, replaceable model interface (QA-01).

6 Peer Review

Candid individual assessment; not shared with other members. Each item graded 1–5.

Evaluation items:

- **Participation** - present and contributing to goals and tasks?
- **Communication** - timely updates; effective use of team tools?
- **Work quality** - claimed backlog tasks; met objectives; added value?
- **Professionalism** - professional, collaborative, taking it seriously?
- **Overall** - overall value to the project this period.

Team member name	Participation	Communication	Work quality	Professionalism	Overall
Ada Lovelace	5	5	5	5	5
Alan Turing	5	4	5	5	5
Grace Hopper	5	5	4	5	5
John Von Neumann	4	4	5	4	4
Charles Babbage	4	3	4	4	4